

Shoot on the Move Theory Proof

Charging Champions - 6560

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1 Theory

By applying kinematics calculations based on odometry readings of the position and velocity of the robot and position of the target, vectors can be calculated and processed to result in a final target vector for velocity and direction of the shoot. This would then be translated through physical constants and some formulas to derive the RPM of the roller-compression system. Additional effects like air resistance and efficiency of spin transfer constants can be calculated post-implementation.

2 Setup

Goal: Get the RPM and turret angle to shoot at while moving at a velocity $\vec{v}$

There are 2 main vectors to keep track of (at the beginning):

let $\vec{r}$ be a "relative position vector", used to track the difference between the robot and target's positions regardless of vertical height

let $\vec{v}$ be a yagsl-determined vector to show the robots velocity in 2 dimensions, also disregarding vertical height

These will not be combined directly, as they are position and velocity, and instead operations will be performed to translate the position vector into a velocity vector.

3 Kinematics

Imagine a triangle in a 2d plane, similar to a standard kinematics projectile motion problem with vectors for velocity and it's components forming a right triangle with an angle θ

Let the vectors on this triangle be made of

$$\vec{l}, \vec{l}_x, \vec{l}_y$$

By solving like a normal projectile motion question, we can separate the vector into componenets and solve from there

$$y(t) = y_0 + \|\vec{l}\| \sin(\theta)t - \frac{1}{2}gt^2$$

And

$$x(t) = \|\vec{l}\| \cos(\theta)t$$

But, drag aplies to both the x and y functions So, using a standard calculation for drag force:

$$F_{drag} = \frac{1}{2} \cdot \rho \cdot A \cdot C_d \cdot v^2$$

Density of air is ≈ 1.225

Contact area = $\pi d = \pi \cdot 0.15m \approx 0.47124$

$$F_{drag} = \frac{1}{2} \cdot 1.225 \cdot 0.47124 \cdot C_d \cdot \vec{l}^2 \approx 0.2886345 \cdot C_d \cdot \|\vec{l}\|^2$$

Substituting into $x(t)$

$$\begin{aligned}
x(t) &= \|\vec{l}\| \cos(\theta) t - \frac{1}{2m} (F_{drag}) t^2 \\
x(t) &= \|\vec{l}\| \cos(\theta) t - \frac{1}{2m} (0.2886345 \cdot C_d \cdot \|\vec{l}\|^2) t^2 \\
2mx &= mt(2\|\vec{l}\| \cos(\theta) - (0.2886345 \cdot C_d \cdot \|\vec{l}\|^2) t) \\
-(0.2886345 C_d \|\vec{l}\|^2) t^2 + 2mt\|\vec{l}\| \cos(\theta) - 2mx &= 0
\end{aligned}$$

(Holy what is this)

$$\|\vec{l}\| = \frac{-2mt \cos(\theta) \pm \sqrt{(-2mt \cos(\theta))^2 - 4(-0.2886345 C_d t^2)(-2mx)}}{2(-0.2886345 C_d t^2)}$$

Using an angle 45° as the value for theta (based on alphabot for now)

$$\begin{aligned}
\|\vec{l}\| &= \frac{-mt\sqrt{2} \pm \sqrt{(-mt\sqrt{2})^2 - 4(-0.2886345 C_d t^2)(-2mx)}}{2(-0.2886345 C_d t^2)} \\
&= \frac{-mt\sqrt{2} \pm \sqrt{2m^2 t^2 - 2.309076 C_d m^2 t^2 x}}{-0.577269 C_d t^2} \\
&= mt \cdot \frac{-\sqrt{2} \pm t(\sqrt{2} - 2.309076 C_d x)}{-0.577269 C_d}
\end{aligned}$$

Take only the + solution because we want the highest arc:

$$= mt \cdot \left(\frac{2.44983458729}{C_d} + \frac{t(\sqrt{2} - 2.309076 C_d x)}{-0.577269 C_d} \right)$$

Drag Coefficients can be determined experimentally, but seeing as we don't have an easy way to experimentally do so, we can estimate it by using Reynold's numbers.

C_d is calculated as a piecewise function of the object's Reynold's number (Re),

$$Re_{fuel} = \frac{\rho \|\vec{l}\| D}{\mu}$$

ρ is the density of air, D is the diameter of the sphere, μ is the viscosity of air.

$$\begin{aligned}
Re_{fuel} &= \frac{1.225 \cdot \|\vec{l}\| \cdot 0.15}{1.8 \cdot 10^{-5}} \\
&\approx 10,208.33 \|\vec{l}\|
\end{aligned}$$

And with exit velocity being in the range from 5 to 20 mph,

$$Re_{fuel} \approx 10,208.33(2.24 \text{ to } 9.94 \frac{m}{s}) \approx 2 \cdot 10^4 \text{ to } 9 \cdot 10^4$$

Since this value is in the subcritical drag range, below drag crisis, C_d can be approximated as 0.47

$$\begin{aligned}
\|\vec{l}\| &= mt \cdot \left(\frac{2.44983458729}{C_d} + \frac{t(\sqrt{2} - 2.309076 C_d x)}{-0.577269 C_d} \right) \approx mt \cdot (5.212414 + \frac{t(\sqrt{2} - 1.08526572x)}{-0.27131643}) \\
&\approx 5.212414mt + \frac{mt^2(\sqrt{2} - 1.08526572x)}{-0.27131643}
\end{aligned}$$

Fuel mass according to the game manual is $m_{fuel} \approx (0.203 \text{ to } 0.227 \text{ kg})$ so can be used as $\approx 0.215 \text{ kg}$

$$\|\vec{l}(x, t)\| \approx 1.12t - 0.8t^2\sqrt{2} - 1.085x$$

Substituting back into $y(t)$

$$y = \frac{\|\vec{l}\|t}{\sqrt{2}} - \frac{(g - \frac{F_{drag}}{m})t^2}{2}$$

$$2\sqrt{2}y = 2t\|\vec{l}\| - \sqrt{2}t^2 \cdot (g - \frac{0.2886345 \cdot C_d \cdot \|\vec{l}\|^2}{m}) \approx 2t\|\vec{l}\| - \sqrt{2}t^2 \cdot (g - 0.630968\|\vec{l}\|^2)$$

Solve for $\|\vec{l}(y, t)\|$

$$-\sqrt{2}t^2 \cdot (g - 0.630968\|\vec{l}\|^2) + 2t\|\vec{l}\| - 2\sqrt{2}y = 0$$

$$\|\vec{l}\| = -\frac{1.122}{t} + \frac{\sqrt{4 + 10.08y + 5.04g}}{1.782t}$$

$$\|\vec{l}(y, t)\| = -\frac{2 + \sqrt{53.4424 + 10.08y}}{1.782t}$$

Then solve for time of flight:

$$-2 - \sqrt{53.4424 + 10.08y} = 1.782t(1.12t - 0.8t^2\sqrt{2} - 1.085x)$$

The nature of this function means it is irreducible, even through Newton-Raphson method. However, time of flight can be implicitly defined as a function of x or y ,

$$t(x, y) = -1.4t^3\sqrt{2} - 1.1x + 2t^2 + \sqrt{53.44 + 10.1y} + 2 = 0$$

When actually running, x and y would be inputted into $t(x, y)$. This would result in a cubic, irreducible function of solely t , meaning Newton-Raphson method can be used to approximate the roots of the function. From there, the value t and the x could be substituted into $\|\vec{l}(x, t)\|$ to finally determine the exit velocity.

4 Vector Processes

Now that there is a vector for magnitude in that plane, we need to work in the ground plane for direction calculations:

First, convert $\vec{l}$'s magnitude to a ground-relative vector, aka $\vec{l}_x$, and call it $\vec{g}$ to store ground velocity of the ball.

$$\|\vec{g}\| = \|\vec{l}\| \cos(\theta) = \frac{x}{t} = x \sqrt{\frac{g}{2(x-y)}}$$

Since the stationary ground velocity of the ball would retain the heading of the position vector $\vec{r}$,

$$\arg(\vec{g}) = \phi = \text{atan2}(\vec{r}_y, \vec{r}_x)$$

This would mean that 0° is upfield and ϕ would increase counter-clockwise from that

Then break it into components:

$$\vec{g} = \|\vec{l}(x, t)\| \cos(\phi) \hat{i} + \|\vec{l}(y, t)\| \sin(\phi) \hat{j}$$

This vector $\vec{g}$ is the field relative target velocity vector.

Next we have to determine the robot's field-relative velocity $\vec{v}$ Given a gyro reading of β , $\vec{v}$ can be calculated

$$\vec{v} = (\vec{v}_x \cos(\beta) - \vec{v}_y \sin(\beta - \frac{\pi}{2})) \hat{i}$$